Mid-Seminar Progress Update: Threat Modeling Tool for 6G NTN

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1. Executive Summary

This update outlines our progress made on the development of a threat modeling tool for a transparent Low Earth Orbit (LEO) Non-Terrestrial Network (NTN) access scenario. The project so far has established a bounded reference architecture, finalized the threat identification methodology using STRIDE, and built a functional web prototype. The primary goal remains the creation of a web application that displays a structured threat database ranked by attack feasibility and impact, accessible via a REST API (Project Plan).

2. Architecture and Data-Flow Overview

The project focuses on a specific LEO scenario connecting a rural user device to an operator's 5G Core. The architecture involves the following key components and data flows:

- **Rural User Device (UE):** the source of the data.
- **Satellite Relay:** a transparent LEO satellite facilitating the link between the UE and ground infrastructure.
- **NTN Gateway:** receives the signal from the satellite and forwards it to the ground.
- **Ground gNB:** the ground-based base station connected to the core.
- **5G Core:** the final destination for the user data.

Trust boundaries: critical trust boundaries exist between the untrusted air interface (UE to Satellite), the satellite link (Satellite to NTN Gateway), and the ground network interfaces (Gateway to gNB/Core). The current architecture documentation details these interfaces and the assets protected within each segment (Project Plan). The detailed Architecture and Data-Flow Diagram, visualizing components, assets and trust boundaries, is attached as **Figure A**.

3. Threat Identification and Ranking Procedures

Identification: we use the **STRIDE** framework (Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege) across all components and flows [4], applied to the six components and five flow groups in a 66-cell review worksheet; as in other STRIDE and risk-matrix studies of LEO satellite systems [6], each threat is then linked to the architecture and rated. **Each threat record stores:** (1) threat actor and attack conditions; (2) assumed security controls; (3) affected assets, components, flows and trust boundaries; (4) attack-feasibility and impact ratings, each with a written reason; (5) source and confidence level; (6) candidate mitigations and their sources.

Ranking: we use **attack feasibility** (F) instead of "likelihood", because the scale measures the access, equipment and skill an attacker needs, not how often an attack happens. The **qualitative priority score = F × I** only orders threats and groups them into bands (Low 1-3, Medium 4-7, High 8-11, Critical 12-16); it is not an exact risk value. The method follows the risk-assessment approach of [5] and is kept deliberately simple.

Feasibility (F) - rated first, for the most realistic attacker, considering the assumed controls: **1 Very low:** exceptional access or resources, rare conditions, little evidence. **2 Low:** specialist equipment or privileged access plus a plausible weakness. **3 Moderate:** commercial equipment or valid credentials and moderate skill; a practical attack path exists. **4 High:** low barrier, commodity tools, repeatable, supported by evidence.

Impact (I): 1 Minor: one UE, quick recovery. **2 Limited:** temporary disruption or limited information exposure. **3 Major:** several users of the shared service, or exposure of sensitive traffic/configuration. **4 Severe:** sustained loss of the selected service or compromise of ground administration.

Why four levels: each level is defined by observable criteria (access, equipment, skill; scope and duration of effect), and four levels avoid a neutral middle value when raters disagree. Every rating needs a one-sentence reason; without evidence the rating is marked as an assumption with Low confidence.

Priority matrix (rows: feasibility, columns: impact)

F ↓ I →	1	2	3	4
4	4	8	12	16
3	3	6	9	12
2	2	4	6	8
1	1	2	3	4

Green Low · Yellow Medium · Orange High · Red Critical

Consistency check: both students independently rate the same 10-15 threat records using the written criteria, record the initial ratings, count agreements for feasibility and impact separately, discuss every difference against the criteria, and document the agreed rating, the reason and the criterion that decided it (comparing and agreeing ratings as in [7]). Unresolved differences are escalated to the supervisor.

4. Status of Prototype and Database

We have moved from the planning phase to a functional prototype. The following deliverables are complete:

- **Web Application:** a functional web app is running in Docker.
- **API Integration:** the application exposes a REST API that supports filtering and ranking of threats, as well as CSV export capabilities.
- **Threat Database:** the database is populated with reviewed threat records.
- **Live Demo:** a live demonstration of the tool within the 6G NTN case study context has been established.

Figure B: not available yet.

5. Example Threat Records

Five completed example records from the database illustrate the STRIDE mapping, ratings with reasons, sources, confidence and mitigations. F = feasibility, I = impact.

ID	Threat; actor and attack conditions	Assumed controls	Mapping; STRIDE	Ratings and reasons; priority	Source; confidence	Candidate mitigation (source)
T-01	UE-Sat link spoofing. Attacker with a radio transmitter inside the beam imitates network signalling.	5G mutual authentication; signalling integrity protection	C1, C2 / A2, A3 / F1 / TB1; **Spoofing**	F2: specialist transmitter and protocol skill, authentication limits success. I3: affects UEs in the area. **2×3 = 6 Medium**	[1],[2]; Low	Mutual authentication; cryptographic signing of access requests [1]
T-02	NTN gateway tampering. Insider or attacker with gateway management access alters configuration.	Admin authentication; configuration backup (assumed)	C3, C4 / A4 / F3, F5 / TB4; **Tampering**	F2: privileged access required. I4: compromise of ground administration. **2×4 = 8 High**	Assumption; Low	TPM-based integrity checks on gateway-to-gNB configuration [4]
T-03	Satellite relay eavesdropping. Passive receiver in the footprint observes radio traffic.	User-plane ciphering active (assumed)	C2 / A1 / F1, F2 / TB1, TB2; **Info. Disclosure**	F3: commodity receiver, no access needed. I2: metadata exposed, ciphertext not decrypted. **3×2 = 6 Medium**	[1],[3]; Medium	Strong encryption of satellite-ground links; frequency hopping [1]
T-04	5G Core interface DoS. Attacker floods signalling or user traffic towards the core path.	Partial rate limiting at core (assumed)	C5 / A3 / F4 / TB3; **DoS**	F2: needs a path past the gateway. I3: several users affected. **2×3 = 6 Medium**	Assumption; Low	Rate limiting and anomaly detection at the gateway [4]
T-05	Administration repudiation. Admin, or attacker with stolen credentials, denies or erases changes.	Local admin logs only (assumed)	C6 / A5 / F5 / TB4; **Repudiation**	F2: privileged access required. I3: changes cannot be traced. **2×3 = 6 Medium**	Assumption; Low	Append-only remote audit log with signed records [4]

6. Individual Contributions

To ensure transparency, the specific contributions of each team member are detailed below:

John:

- Defined the bounded reference architecture and trust boundaries in coordination with the 3GPP NTN baseline.

- Developed the STRIDE-based threat identification framework.
- Authored the initial threat records and scoring logic in the database.
- Documented the intermediate risk analysis.

Marius:

- Designed and implemented the functional web application prototype.
- Developed the REST API to serve the threat database.
- Implemented the ranking algorithms and user-interface filters.
- Managed the Docker deployment and integration with Open WebUI.

7. Mid-Seminar Plan (21 October)

The mandatory mid-seminar is scheduled for **21 October 2026** (Project Team Charter). Our plan for the presentation includes:

1. **Demonstration:** a live walkthrough of the functional web application, showcasing the threat database, the STRIDE filtering capabilities, and the ranking interface.
2. **Architecture Review:** presentation of the detailed architecture and data-flow diagram.
3. **Risk Discussion:** presentation of the identified threats, their feasibility/impact ratings and priority scores, and the specific rationale (evidence or assumptions) behind them.
4. **Method and Consistency Check:** presentation of the feasibility × impact method, the priority matrix and preliminary agreement results from the independent rating of 10-15 threats.
5. **Proof of Completeness:** verification that both students can independently install and run the core application using documented setup instructions.

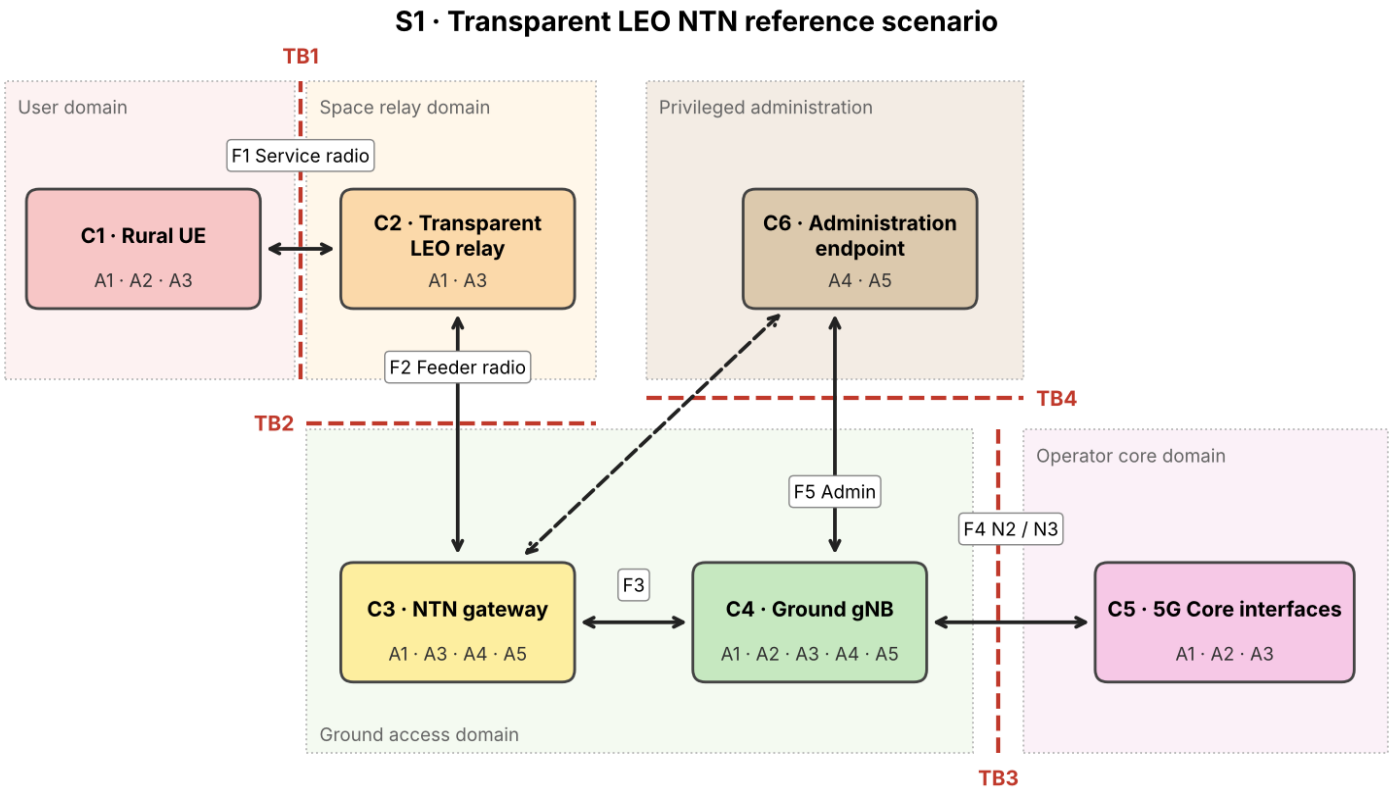
8. Current Issues and Blockers

- **Reference Consolidation:** while the core scenarios are based on 3GPP standards, the final bibliographic references are still being consolidated into the threat database.
- **Consistency Check Timing:** the independent rating of 10-15 records and the documentation of resolved differences must be finished before the seminar to give preliminary results; the full evaluation follows on 22-28 October.
- **Evidence Confidence:** most example records currently have Low evidence confidence; section-level source checks are still to be completed.
- **Assumption Labeling:** some threat scores currently rely on assumptions due to the lack of specific ground infrastructure data. We are actively labeling these assumptions to maintain transparency in the final report.

References

[1] 3GPP TS 33.501, “Security architecture and procedures for 5G system.” [2] 3GPP TR 38.821, “Solutions for NR to support non-terrestrial networks (NTN).” [3] J. Pavur et al., “A Tale of Sea and Sky: On the Security of Maritime VSAT Communications,” IEEE S&P, 2020. [4] A. Shostack, *Threat Modeling: Designing for Security*, Wiley, 2014. [5] Ansong et al., “Role of Cybersecurity for a Secure Global Communication Eco-System,” 2025, doi:10.1016/j.cose.2024.104156. [6] Frøseth et al., “Threat Modeling in Satellite Communications for Maritime Operations,” 2025, doi:10.1007/978-3-031-82349-7_26. [7] Suomalainen et al., “Cybersecurity for Tactical 6G Networks,” 2025, doi:10.1016/j.future.2024.107500. Project Plan and Project Team Charter, D7086E, Luleå University of Technology, 2026.

Figure A - Architecture and data-flow diagram (S1)



C: components A: assets (A1 user traffic C/I, A2 subscriber identity/auth, A3 service availability, A4 config & admin credentials, A5 admin records/logs)
F: flows (F5 dashed = admin path to C3) TB: trust boundaries (dashed red)

Flows: F1 service radio (C1-C2, TB1); F2 feeder radio (C2-C3, TB2); F3 ground transport (C3-C4); F4 N2/N3 core connection (C4-C5, TB3); F5 administration (C6 to C3/C4, TB4). Assets: A1 confidentiality/integrity of user traffic; A2 subscriber identity and authentication material; A3 service availability and access capacity; A4 ground configuration and administration credentials; A5 administration change records and logs.